Himavanth Karpurapu

695 S 11th St, San Jose, CA 95112 | himvanth89@gmail.com | 408-210-9874 | LinkedIn | GitHub | Portfolio

Awards

AWS re:Invent All Builders Welcome Grant Scholar (100 selected globally) · SJSU Grad Slam 2026 Winner · TreeHacks 2026 Healthcare AI Track Winner (Stanford) · LA Hacks 2026 First Place · Shopify Shop Mini Winner · SpartUp 2026 Winner · Mozilla Responsible AI Winner · UC Berkeley AI Hackathon Winner · YC Startup School 2026

Education

San Jose State University, M.S. in Computer Science; GPA: 4.0 / 4.0 — Aug 2024 – Dec 2026

Jawaharlal Nehru Technological University, B.Tech in Computer Science — Aug 2019 – May 2023

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, C, C++, SQL, Go

LLM and Agents: LangGraph, LangChain, LlamaIndex, MCP, Fetch.ai uAgents, Hugging Face, prompt engineering, agentic workflows

RAG and ML: OpenSearch, Pinecone, Neo4j, BM25, hybrid search, vector databases, PyTorch, TensorFlow

Full Stack: React, Node.js, Express, Django, Flask, FastAPI, Spring Boot, React Native, REST APIs, WebSockets, JWT

Cloud: Amazon Web Services / AWS (Lambda, API Gateway, S3, DynamoDB, ECS, ECR, EC2, VPC, IAM, CloudWatch, Bedrock)

Databases: PostgreSQL, MySQL, DynamoDB, MongoDB, Redis

DevOps: Docker, Kubernetes, CircleCI, GitHub Actions, Git, Linux, CI/CD, Agile, JIRA

Experience

Research Intern , San Jose State University (Advisor: Dr. Robert Chun) — May 2025 – Present

- Formalized the Generation Verification Gap in LLMs as G(M,T) minus V(M,T) using LangGraph, designing evaluation pipelines that quantify conditions where models solve problems but cannot verify their own outputs.
- Built and tested pipelines across GPT 4o, Claude 3.5 Sonnet, Llama 3.1, and DeepSeek on GSM8K, MATH, HumanEval, and MBPP, analyzing failure modes including false positives, calibration errors, and diversity collapse.
- Won the SJSU Grad Slam 2026 presenting findings on when LLMs recognize their own mistakes.

Software Engineer , WinIT, Hyderabad, India — May 2022 – Jul 2024

- Delivered features for a production platform on AWS Lambda, API Gateway, and DynamoDB, using GitHub Actions and Agile sprints across the full software development life cycle.
- Wrote and maintained Lambda functions and SQL queries, improving performance on high traffic endpoints.
- Implemented JWT authentication in Java with Kubernetes secrets and led security reviews that strengthened access controls.

Projects

Rewind , Multi Agent Scheduling System (TreeHacks 2026 Healthcare AI Track Winner) — GitHub

- Designed and built a 7 agent autonomous scheduling system using Fetch.ai uAgents and Agentverse, with specialized agents coordinating calendar, email, tasks, and user context, plus Meta Ray-Ban smart glasses for hands free voice capture. Stack: Python, Fetch.ai uAgents, OpenAI API.

Dispatch , Local First Multimodal File Index (LA Hacks 2026 First Place) — GitHub | Devpost

- Built a local first retrieval engine indexing files across text, image, and code using SQLite FTS5, LanceDB, SigLIP, and Nomic embeddings, with an MCP server for tool integration. Stack: Electron, Node.js, llama.cpp (Gemma 3 4B).

NeuroBoost , AI Startup (SpartUp 2026 Winner, YC Startup School 2026) — GitHub

- Built an AI productivity platform (React, Django and Node.js, PostgreSQL) serving roughly 1,000 users, with LLM features including a LangChain pipeline turning voice notes into tasks and a priority graph scheduler; iterated across three product pivots based on user research with 120 students.

Scalable Three Tier Web Application Migration to AWS , Cloud and DevOps — GitHub

- Architected a three tier application (React, Node.js and Express, MongoDB) containerized on AWS ECS and ECR with auto scaling and CloudWatch monitoring, plus a CircleCI CI/CD pipeline gated on Jest tests and security across VPC, IAM, and CORS.

Certifications and Leadership

AWS Solutions Architect · AWS Machine Learning Specialty · AWS Generative AI Developer · CodePath Technical Mentor (1,000+ students) · Adobe Campus Ambassador · Claude Campus Ambassador · Perplexity Campus Ambassador